VERIQTA SUBSCRIBER SERIES #4

VERIQTA CASE FILE #004
THE TERRAFORM CHANGE THAT BROKE EVERYTHING

FROM CHAOS TO CONTROL.

REAL MISTAKES. REAL IMPACT.
REAL SOLUTIONS THAT WORK.

NO PLAN
Nobody reviewed the plan.

WRONG VARIABLE
One variable changed everything.

STATE DRIFT
Terraform lost track of reality.

NO PROTECTION
Production had no guardrails.

NO ROLLBACK
Recovery took too long.

NO INVESTIGATION
We flew blind. We guessed.

SAFE ARCHITECTURE
Designed for safety, visibility, and control.

ENVIRONMENT GUARDS
Protected by layers. No direct access.

OBSERVABILITY
Full visibility. Better insights.

SMART ALERTS
Real-time signals. Faster response.

CONTINUOUS IMPROVEMENT
Learn, adapt, and get better.

CONFIDENCE
Safe changes. Reliable systems. Happy customers.

TERRAFORM IS POWERFUL.
BUT WITHOUT PROCESS, VISIBILITY, AND PROTECTION, IT CAN BREAK YOU.

BUILD IT RIGHT.
RUN IT SAFE.
SLEEP AT NIGHT.

We didn't just fix our Terraform.
We fixed the way we build.

CODE TO PRODUCTION SAFELY | AUTOMATE WITH GUARDRAILS | PROTECT EVERY ENVIRONMENT | OBSERVE CONTINUOUSLY | IMPROVE EVERY DAY

VERIQTA SUBSCRIBER SERIES #

GOOD ENGINEERS WRITE CODE. GREAT ENGINEERS BUILD SYSTEMS THAT DON'T BREAK.

VERIQTA

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VERIQTA CASE FILE #004

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PAGE 1

How One Infrastructure Change Destroyed an Entire Environment

1. INCIDENT OVERVIEW

Friday,
4:42 PM

A routine Terraform deployment started.

The pull request had been approved.

The pipeline was green.

Terraform completed successfully.

- [x] No errors.
- [x] No warnings.
- [x] No failed resources.

FIVE MINUTES LATER...

- Monitoring alerts started firing.
- Internal services became unreachable.
- Customer traffic dropped sharply.
- Engineers declared a production incident.

The deployment succeeded.
The infrastructure didn't.

2. INCIDENT SNAPSHOT

USERS AFFECTED
23,000+

SERVICES IMPACTED
12

DOWNTIME
1 Hour 23 Minutes

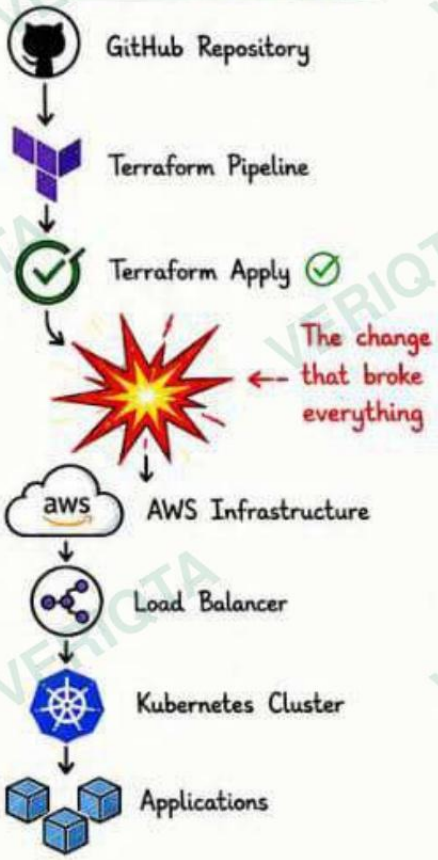
SEVERITY
Critical

BUSINESS IMPACT
High

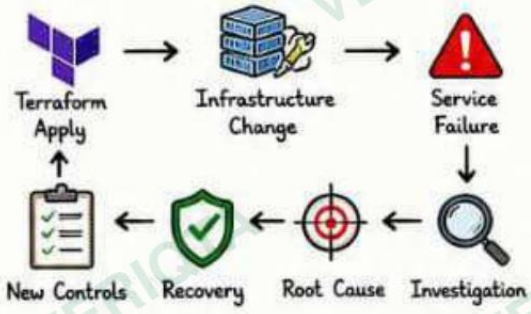
LESSON #1

Successful automation does not guarantee successful outcomes.
Tools execute instructions.
They do not validate business impact.

3. ARCHITECTURE SKETCH



4. INCIDENT FLOW (MIND MAP)



5. WHAT MADE THIS INCIDENT DIFFICULT?

Both statements were true.

- [x] Terraform reported: APPLY COMPLETE
- [ ] Production reported:
  - Services Unavailable
  - Traffic Failing
  - Customer Impact

6. INTERVIEW NOTE

Q: Why is Terraform considered powerful and dangerous at the same time?

A: Because a single change can modify hundreds of infrastructure resources within minutes.

7. PROGRESS TRACKER

1 INCIDENT OVERVIEW · 2 NO PLAN REVIEW · 3 WRONG VARIABLE · 4 STATE FILE PROBLEMS · 5 NO ENV PROTECTION · 6 NO ROLLBACK STRATEGY · 7 INVESTIGATION TIMELINE · 8 NEW TERRAFORM ARCHITECTURE · 9 READINESS CHECKLIST · 10 10 TERRAFORM LESSONS

"Infrastructure as Code can scale your success.
It can also scale your mistakes."

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SUBSCRIBER SERIES #

MISTAKE #1

NOBODY REVIEWED THE PLAN

We trusted the apply.
We ignored the plan.

THE PLAN SHOWS YOU WHAT WILL HAPPEN BEFORE IT ACTUALLY HAPPENS.

DON'T SKIP IT.

WHAT HAPPENED?

A large Terraform change was raised to update the infrastructure.
The plan output was huge.
No one reviewed it.
The apply was executed.
Terraform made all the changes exactly as written.

THE PLAN OUTPUT (EXAMPLE)

```
$ terraform plan
...
Plan: 542 to add, 128 to change,
32 to destroy.
...
# aws_security_group.web will be replaced
# aws_lb.app will be recreated
# aws_subnet.private[0] will be destroyed
# aws_db_instance.main will be replaced
...
```

Too many changes. Too much risk.
But no one reviewed.

WHY THIS WAS DANGEROUS

- Critical resources were being destroyed or recreated.
- Dependencies were not understood.
- Changes happened faster than anyone could react.
- The plan output was treated like a formality, not a safety net.

A plan is a prediction of the future.
Ignoring it is like deploying blind.

Terraform

LESSON LEARNED

- [x] Always review every plan.
- [x] Understand what will change.
- [x] Ask questions for any unexpected change.
- [x] Don't trust. Validate.
- [x] Make plan review a mandatory step.

INTERVIEW QUESTION

Q: Why is terraform plan important?

A: It shows exactly what changes Terraform will make to your infrastructure before those changes are actually applied.

BAD HABIT TO AVOID

- Running apply without plan review.
- Glancing at the plan output.
- Ignoring warnings in the plan.
- Assuming automation means safety.

VERIQTA CASE FILE #004
THE TERRAFORM CHANGE THAT BROKE EVERYTHING

MISTAKE #2

WRONG VARIABLE. ENTIRE ENVIRONMENT.

One small change in a variable caused a massive infrastructure impact.

VARIABLES ARE POWERFUL. TREAT THEM LIKE CODE.

ONE WRONG VALUE CAN BREAK PROD.

WHAT HAPPENED?

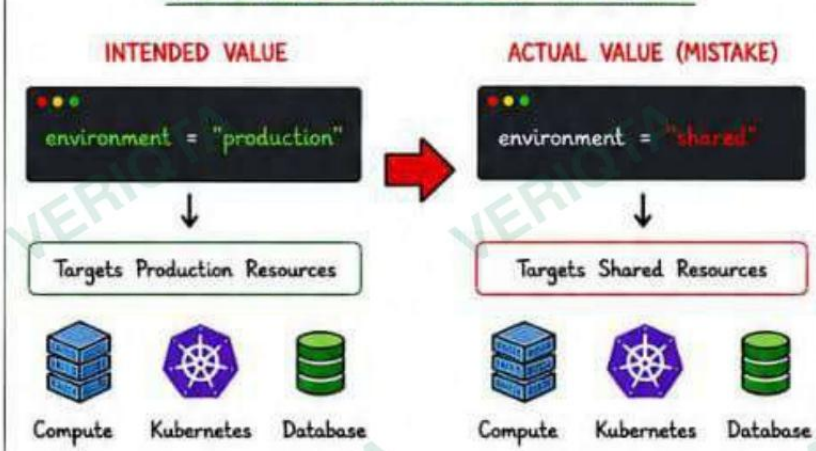
A variable that defines the environment was changed accidentally.

Instead of targeting production,

Terraform targeted the shared environment.

Resources were destroyed and recreated across the board.

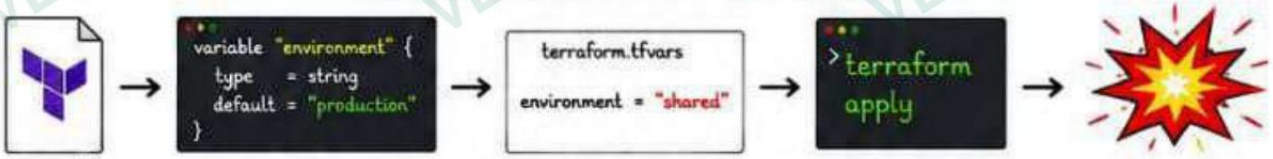
THE VARIABLE THAT CAUSED THE CHAOS



IMPACT

- Resources Recreated
- Services Went Down
- Traffic Interrupted
- Business Impact

HOW ONE VARIABLE CHANGED EVERYTHING



WHY DID THIS HAPPEN?

- [x] No validation on variable values.
- [x] No environment restrictions.
- [x] Variables file was not reviewed.
- [x] Assumed value was correct.
- [x] Lack of second pair of eyes.

Assumption is the enemy of production.

INTERVIEW QUESTION

Q: Why is variable validation important in Terraform?

A: Because it prevents incorrect or unexpected values from being applied, which can cause unintended changes to critical infrastructure.

BEST PRACTICES

- [x] Validate all critical variables.
- [x] Restrict values where possible.
- [x] Separate variable files per environment.
- [x] Review variables before every apply.
- [x] Use automated checks in CI/CD.

PROGRESS TRACKER

1 INCIDENT OVERVIEW → 2 NO PLAN REVIEW → 3 WRONG VARIABLE → 4 STATE FILE PROBLEMS → 5 NO ENV PROTECTION → 6 NO ROLLBACK STRATEGY → 7 INVESTIGATION TIMELINE → 8 NEW TERRAFORM ARCHITECTURE → 9 READINESS CHECKLIST → 10 10 TERRAFORM LESSONS

Variables are not just values. They are decisions.
Make the right decision, every time.

MISTAKE #3

TERRAFORM LOST TRACK OF REALITY

The state file and reality were no longer the same.

STATE IS THE SOURCE OF TRUTH.
PROTECT IT.
MANAGE IT.

WHAT HAPPENED?

Terraform uses the state file to know what exists.

But the state file became out of sync with reality.

Terraform thought resources were one thing.

Reality was something else.

This caused wrong actions during apply.

STATE FILE vs REAL INFRASTRUCTURE

TERRAFORM STATE
(What it thinks)

terraform.tfstate

```
{
  "aws_instance.web": {
    "id": "i-0abcd1234efgh5678",
    "status": "running"
  }
}
```

≠

REAL INFRASTRUCTURE
(What actually exists)

⚠ When state and reality drift, Terraform makes dangerous decisions.

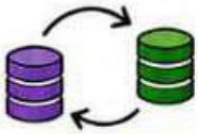
IMPACT

- Resources Destroyed
- Services Went Down
- Unexpected Changes
- High Business Impact

WHY STATE DRIFTS

MANUAL CHANGES
Changes made outside of Terraform.

STATE FILE CORRUPTION
State file became corrupted.



MISSING STATE LOCKING
Multiple applies overwrote state.

DELETED RESOURCES OUTSIDE TF
Resources deleted manually.

MODIFIED TF CONFIG
Configuration changed without proper plan.

HOW TO PREVENT STATE PROBLEMS

- ☑ Use remote state (S3 + DynamoDB lock).
- ☑ Enable state locking to prevent concurrent applies.
- ☑ Never edit state file manually.
- ☑ Detect and fix drift regularly.
- ☑ Review state before every major change.

INTERVIEW QUESTION

Q: What happens if the Terraform state file is lost or corrupted?

A: Terraform will not know what exists. It may try to create duplicate resources or destroy the wrong ones.

LESSON LEARNED

State is the memory of your infrastructure.

If memory is wrong, decisions will be dangerous.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW ✓	NO PLAN REVIEW ✓	WRONG VARIABLE ✓	STATE FILE PROBLEMS	NO ENV PROTECTION	NO ROLLBACK STRATEGY	INVESTIGATION TIMELINE	NEW TERRAFORM ARCHITECTURE	READINESS CHECKLIST	10 TERRAFORM LESSONS

"If the state is wrong, everything Terraform does will be wrong."

VERIQTA CASE FILE #004
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MISTAKE #4

NO ENVIRONMENT PROTECTION.

PRODUCTION WAS TOO EASY TO CHANGE.

PRODUCTION NEEDS GUARDS. NOT JUST PERMISSIONS. IT NEEDS PROTECTIONS.

WHAT HAPPENED?

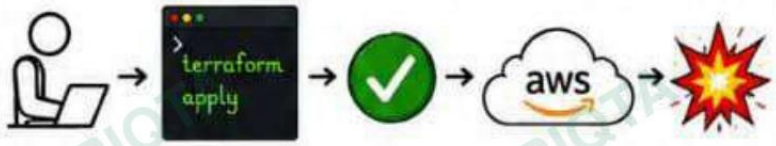
Any engineer with access could apply changes directly to production.

No approval.
No checks.
No safety nets.

One click was all it took.

Production was wide open.

THE PATH TO PRODUCTION (TOO EASY)



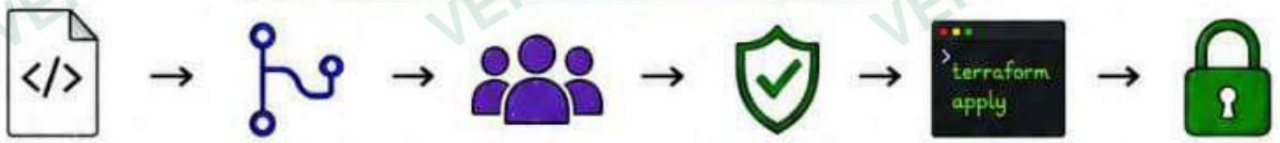
Engineer Makes Change → Runs Terraform Apply → Apply Succeeds → Changes Applied to Production → Incident Happens

⚠ No approvals. No checks. No controls.
Too easy to impact production.

IMPACT

- Unauthorized Changes
- Faster Time to Break
- Higher Risk
- Business Impact

WHAT ENVIRONMENT PROTECTION SHOULD LOOK LIKE



Code Change Committed → Pull Request Created → Review & Approval → Policy Checks (Automated) → Apply to Production → Protected Production

✓ Add gates. Add checks. Add approvals. Protect what matters.

WHY PROTECTION MATTERS

- ☑ Prevents accidental changes.
- ☑ Ensures accountability.
- ☑ Enforces best practices.
- ☑ Adds time to think.
- ☑ Reduces the blast radius.

⚠ Speed is good. Controlled speed is better.

INTERVIEW QUESTION

Q: Why is environment protection important in Terraform?

A: Because it prevents unauthorized or accidental changes to critical environments like production. It adds safety, accountability, and control.

BEST PRACTICES

- ☑ Require PR and code review.
- ☑ Use approval workflows for prod.
- ☑ Enforce policy checks.
- ☑ Use separate workspaces.
- ☑ Apply least privilege access.
- ☑ Log and audit all changes.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW ✓	NO PLAN REVIEW ✓	WRONG VARIABLE ✓	STATE FILE PROBLEMS ✓	NO ENV PROTECTION	NO ROLLBACK STRATEGY	INVESTIGATION TIMELINE	NEW TERRAFORM ARCHITECTURE	READINESS CHECKLIST	10 TERRAFORM LESSONS

"The easiest path to production is the most dangerous path."

VERIQTA CASE FILE #004
THE TERRAFORM CHANGE THAT BROKE EVERYTHING

MISTAKE #5

NO ROLLBACK STRATEGY. NO EXIT PLAN.

EVERY DEPLOYMENT NEEDS A WAY BACK.
PLAN FOR FAILURE BEFORE IT HAPPENS.

We could deploy changes.
But we couldn't undo them.

WHAT HAPPENED?

The apply changed critical infrastructure.

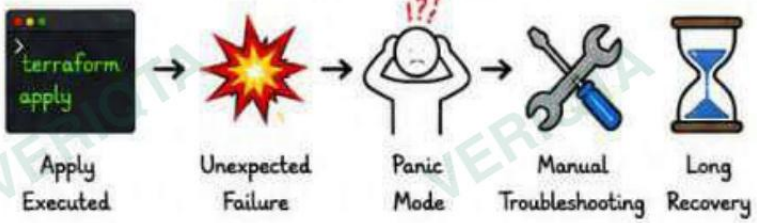
When things broke, we had no rollback plan.

Terraform has no built-in rollback.

We had to manually fix everything.

Recovery took longer than the outage.

THE PROBLEM WITH NO ROLLBACK



No rollback. No automation.
Only manual fixes and guesswork.

IMPACT

- Longer Downtime
- Slower Recovery
- Higher Risk
- Business Impact

WHAT A ROLLBACK STRATEGY LOOKS LIKE

Good engineers plan deployment. Great engineers plan recovery.

WHY ROLLBACK MATTERS

- ☑ Failures are inevitable.
- ☑ Rollback reduces downtime.
- ☑ Protects customers and business.
- ☑ Builds confidence in changes.
- ☑ A safety net encourages better changes.

⚠ Without rollback, every change is a gamble.

INTERVIEW QUESTION

Q: Does Terraform have a built-in rollback?

A: No. Terraform maintains the desired state. It cannot automatically rollback. You must design your own rollback strategy using backups, versioning, or blue/green deployments.

BEST PRACTICES

- ☑ Keep state backups.
- ☑ Use version control for all changes.
- ☑ Test changes in lower environments.
- ☑ Document rollback steps.
- ☑ Use blue/green or canary strategies for critical infra.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW ✓	NO PLAN REVIEW ✓	WRONG VARIABLE ✓	STATE FILE PROBLEMS ✓	NO ENV PROTECTION ✓	NO ROLLBACK STRATEGY ✓	INVESTIGATION TIMELINE	NEW TERRAFORM ARCHITECTURE	READINESS CHECKLIST	10 TERRAFORM LESSONS

"You don't rise to the level of your goals.
You fall to the level of your systems."

VERIQTA CASE FILE #004
THE TERRAFORM CHANGE THAT BROKE EVERYTHING

PAGE 7

MISTAKE #6

NO INVESTIGATION TIMELINE.

We were flying blind.
We didn't know what broke, when, or why.

YOU CAN'T FIX WHAT YOU CAN'T SEE.
INVESTIGATE FIRST.

WHAT HAPPENED?

After the apply, production broke.

But we had no timeline.

No visibility.

No clue what changed, what failed, or what triggered the outage.

We lost time because we had no investigation process.

We fixed blindly.
We guessed.

THE INVESTIGATION TIMELINE (WHAT ACTUALLY HAPPENED)

10:02 AM
Terraform apply executed.
Apply succeeded.

10:03 AM
Services started failing.
Errors in multiple systems.

10:04 AM
Traffic dropped.
Alerts triggered.

10:05 AM
Engineers started investigating.
No idea what changed.

10:20 AM
Found out wrong resources were changed. State was out of sync.

11:15 AM
Started manual recovery.
Services restored gradually.

⚠ Over 1 hour of confusion, delay, and impact.
All because we had no investigation timeline.

IMPACT

- Longer Downtime
- Slower Root Cause Discovery
- Wrong Fixes Tried
- Higher Business Impact
- Loss of Customer Trust

WHAT A GOOD INVESTIGATION TIMELINE PROVIDES

Capture Every Event → Record Exact Time → Understand What Happened → Find Root Cause → Learn & Improve → Prevent Next Time

✓ A timeline brings clarity. Clarity brings faster recovery.

LESSON LEARNED

- ☑ Every incident needs a timeline.
- ☑ Capture facts, not assumptions.
- ☑ Time is critical in investigations.
- ☑ Visibility reduces downtime.
- ☑ Document everything.

⚠ If it's not recorded, it didn't happen.

INTERVIEW QUESTION

Q: Why is an investigation timeline important in an incident?

A: It helps us understand what happened, when it happened, and why it happened. It reduces downtime, prevents wrong fixes, and helps avoid the same issues in the future.

BEST PRACTICES

- ☑ Start timeline the moment something changes.
- ☑ Record every action and observation.
- ☑ Use incident channels for updates.
- ☑ Include screenshots and logs.
- ☑ Review timeline in postmortem.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW ✓	NO PLAN REVIEW ✓	WRONG VARIABLE ✓	STATE FILE PROBLEMS ✓	NO ENV PROTECTION ✓	NO ROLLBACK STRATEGY ✓	INVESTIGATION TIMELINE	NEW TERRAFORM ARCHITECTURE	READINESS	10 TERRAFORM

Investigate with discipline. Recover with confidence.
Facts first. Fix next.

SOLUTION #1

SAFE TERRAFORM ARCHITECTURE

We redesigned our Terraform workflow.
Safety, visibility, and control at every step.

GOOD ARCHITECTURE PREVENTS INCIDENTS.

DESIGN FOR SAFETY, NOT JUST SPEED.

WHAT WE BUILT

We built a safe, controlled, and observable Terraform architecture.

Every change is reviewed, validated, and approved before it touches production.

No more blind applies.
No more surprises.

THE SAFE TERRAFORM ARCHITECTURE

1. CODE COMMIT
Engineer commits Terraform code.

2. PLAN & VALIDATE
Terraform plan is generated and validated.

- tflint
- terraform validate
- policy checks

3. PEER REVIEW
Team reviews the plan and changes.

4. APPROVAL GATE
Required approvals before applying changes.

5. APPLY (CONTROLLED)
Changes applied with confidence and logging.

6. MONITOR & OBSERVE
Monitor, alert, and validate after apply.

Every change is planned. Every change is reviewed. Every change is safe.

KEY SAFETY CONTROLS

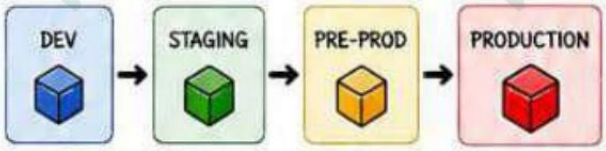
STATE LOCKING
Prevent concurrent applies and state corruption.

POLICY ENFORCEMENT
Use Sentinel/OPA to enforce rules and compliance.

LEAST PRIVILEGE
Terraform runs with minimal permissions.

VERSION CONTROL
All changes are tracked, reviewed, and auditable.

ENVIRONMENTS & PROMOTION FLOW



Safe to Experiment → Validate Everything → Final Checks & Approval → Controlled Deployment

Promote the same code through environments.
No direct changes in production.

IMPACT

- Unauthorized Changes Blocked
- Faster Safe Deployments
- Lower Risk of Failure
- Better Business Continuity
- Higher Customer Trust

WHY THIS MATTERS

- ☑ Prevents accidental changes.
- ☑ Provides visibility and auditability.
- ☑ Catches issues before production.
- ☑ Enforces consistency and standards.
- ☑ Builds confidence in deployments.

Good architecture is your best incident response.

INTERVIEW QUESTION

Q: What is the most important part of a safe Terraform architecture?

A: The process. Code is important, but process (review, validation, and approvals) is what prevents disasters.

BEST PRACTICES

- ☑ Keep Terraform code in Git.
- ☑ Always run plan and review.
- ☑ Enforce policies automatically.
- ☑ Require approvals for production.
- ☑ Monitor and log every apply.
- ☑ Continuously improve the process.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW ✓	NO PLAN REVIEW ✓	WRONG VARIABLE ✓	STATE FILE PROBLEMS ✓	NO ENV PROTECTION ✓	NO ROLLBACK STRATEGY ✓	INVESTIGATION TIMELINE	NEW TERRAFORM ARCHITECTURE	READINESS CHECKLIST	10 TERRAFORM LESSONS

Good architecture prevents incidents.
Safe processes prevent disasters.

SOLUTION #2

ENVIRONMENT GUARDS & PROTECTIONS.

PRODUCTION IS PRECIOUS.
WE PROTECT IT WITH LAYERS.
SAFETY BY DESIGN.

We added strong protections between code and production.
No more direct access. No more risky changes.

WHAT WE BUILT

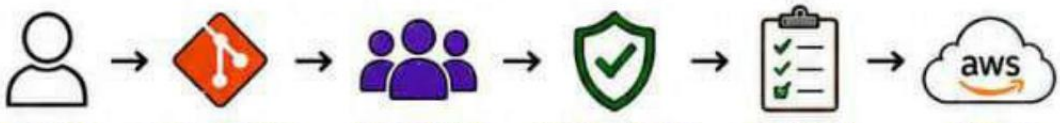
We created guardrails that stop risky changes before they reach production.

Every environment has its own rules.

Every action has controls.

Protected pipelines.
Protected infrastructure.
Protected business.

ENVIRONMENT PROTECTIONS OVERVIEW



ENGINEER — Makes change in safe way.

PULL REQUEST — Change requires review.

CODE REVIEW — Team reviews for safety.

POLICY CHECKS — Automated policy validation.

APPROVAL — Required approval to proceed.

DEPLOY — Change deployed safely.

No direct changes. No bypass. No risk to production.

PROTECTION LAYERS

ACCESS CONTROL
No direct production access. Everything via pipeline.

SENTINEL / OPA POLICIES
Enforce rules and compliance. Block unsafe changes.

REQUIRED APPROVALS
Human approval required for sensitive changes.

ENVIRONMENT ISOLATION
Separate state, accounts, and permissions.

POLICY EXAMPLES

```
# Block public S3 buckets
main = rule {
  all aws_s3_bucket as b {
    b.acl != "public-read"
  }
}

# Require tagging
main = rule {
  all resources as r {
    r.tags.env is not empty
} }
```

Policies catch problems.
Humans catch context.
Together we stay safe.

IMPACT

- Unauthorized Changes Prevented
- Faster & Safer Deployments
- Lower Risk of Outages
- Better Compliance & Auditability
- Higher Customer Confidence

WHY THIS MATTERS

- ☑ Stops risky changes early.
- ☑ Enforces standards automatically.
- ☑ Protects production environments.
- ☑ Reduces human error.
- ☑ Builds trust and confidence.

Guardrails don't slow you down.
They keep you out of trouble.

INTERVIEW QUESTION

Q: Why are environment protections critical in Terraform?

A: Because they prevent unauthorized changes, enforce policies, require approvals, and separate environments. They reduce risk and ensure safe, controlled deployments.

BEST PRACTICES

- ☑ Use least privilege access.
- ☑ Enforce policies with Sentinel/OPA.
- ☑ Require code review and approvals.
- ☑ Isolate environments completely.
- ☑ Continuously audit and improve.
- ☑ Monitor policy violations.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW	NO PLAN & REVIEW	WRONG VARIABLE	STATE FILE PROBLEMS	NO ENV PROTECTION	NO ROLLBACK STRATEGY	INVESTIGATION TIMELINE	SAFE TERRAFORM ARCHITECTURE	ENV GUARDS & PROTECTIONS	10 TERRAFORM LESSONS
✓	✓	✓	✓	✓	✓	✓	✓	✓	

"Protect your environments like your business depends on it.
Because it does.

SOLUTION #3

OBSERVABILITY & CONTINUOUS IMPROVEMENT.

We don't stop at deployment.
We continuously monitor, learn, and improve.

OBSERVABILITY TURNS CHANGES INTO INSIGHTS.
IMPROVEMENT TURNS INSIGHTS INTO SAFETY.

WHAT WE BUILT

We implemented full observability across the Terraform lifecycle.

We measure everything that matters.

We learn from every change we make.

We improve continuously.

From incidents to insights.
From insights to safety.

OUR OBSERVABILITY FRAMEWORK

1. PLAN VISIBILITY
Capture plan details, impact, and intent.

2. APPLY VISIBILITY
Track applies, approvals, and who/what/when.

3. RESOURCE VISIBILITY
Monitor resources, drift, health, and configuration.

4. ALERTING & NOTIFICATIONS
Real-time alerts for failures, drift, and anomalies.

5. METRICS & DASHBOARDS
Visualize trends, changes, and environment health.

6. FEEDBACK & IMPROVEMENT
Review, learn, and improve every iteration.

✓ Full visibility. Early signals. Faster response. Continuous improvement.

WHAT WE MONITOR

- **PLAN METRICS** — Resources to be changed, add/update/delete counts.
- **APPLY METRICS** — Success/failure rates, duration, and frequency.
- **RESOURCE HEALTH** — State drift, config changes, and compliance status.
- **ALERTS** — Failures, drifts, policy violations, and risk signals.
- **COST & USAGE** — Cost impact, usage trends, and optimization opportunities.
- **PEOPLE & PROCESS** — Approvals, reviews, and team performance.

SAMPLE DASHBOARD VIEW

Measure what matters. See what's changing.
Improve what we control.

IMPACT

- Fewer Incidents
- Faster Detection & Response
- Lower Risk of Failures
- Reduced Cost & Waste
- Higher Team Confidence
- Better Decisions with Data

WHY THIS MATTERS

- ☑ Observability reduces unknowns.
- ☑ Early signals prevent big outages.
- ☑ Data drives better decisions.
- ☑ Continuous improvement builds resilient systems.

⚠ If you can't see it, you can't fix it.

INTERVIEW QUESTION

Q: What is the goal of observability in Terraform?

A: To continuously monitor and understand the behavior of our infrastructure and processes so we can detect issues early, respond faster, and continuously improve.

BEST PRACTICES

- ☑ Log every plan and apply.
- ☑ Monitor drift continuously.
- ☑ Set meaningful alerts.
- ☑ Build dashboards for visibility.
- ☑ Review metrics regularly.
- ☑ Turn insights into improvements.

PROGRESS TRACKER

1	2	3	4	5	6	7	8	9	10
INCIDENT OVERVIEW	NO PLAN REVIEW	WRONG VARIABLE	STATE FILE PROBLEMS	NO ENV PROTECTION	NO ROLLBACK STRATEGY	INVESTIGATION TIMELINE	SAFE TERRAFORM ARCHITECTURE	ENV GUARDS & PROTECTIONS	OBSERVABILITY & IMPROVEMENT
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

"Good systems prevent problems.
Great systems learn and improve.
Keep observing. Keep learning. Keep getting better."